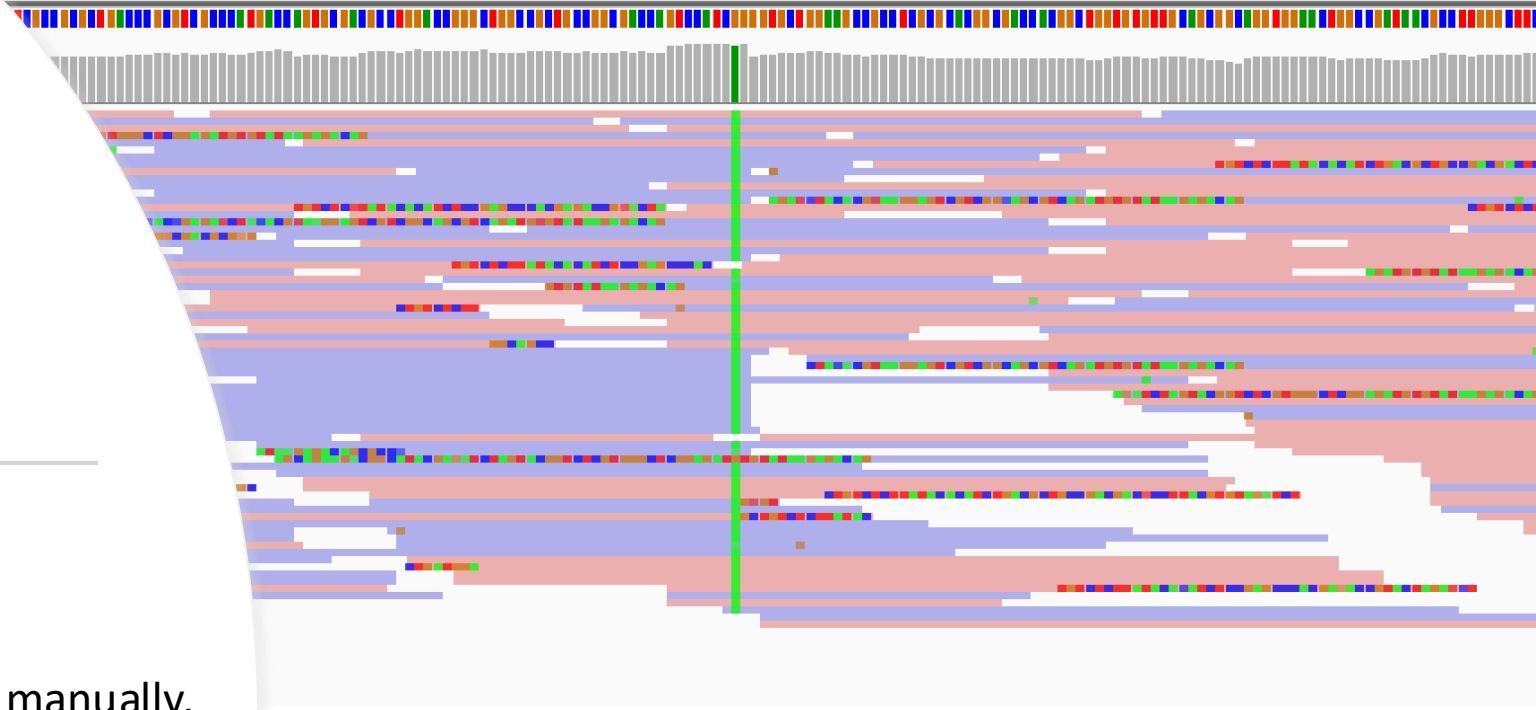
BIO392 : Seqeunce analysis introduction

Sequence QC and Alignment

Worawich Phornsiricharoenphant

Why do you have to do bioinformatics analysis?



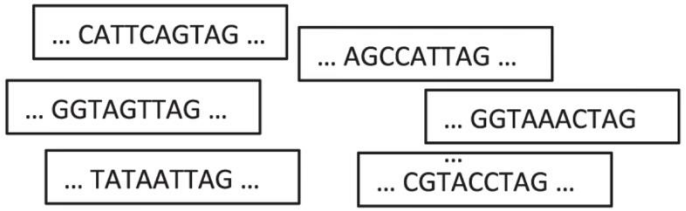
- **Scale of Data** : Impossible to find genetic variant manually.
- **Interpretation** : Needed tool to interpret sequence of A, T, C and G into insightful information.
- **Integration** : Combining data from different experiments to build a more complete picture of a biological system.

https://blogs.technet.microsoft.com/

Ways to do it

Genomic DNA

Next-generation DNA sequencing



millions-billions of ***reads***
~30-1000 nucleotides

Resequencing

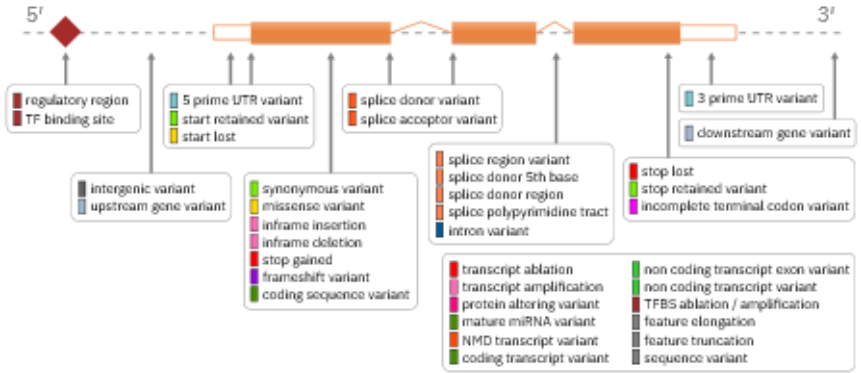
Align reads to *reference genome* and identify variants

***De novo* assembly**

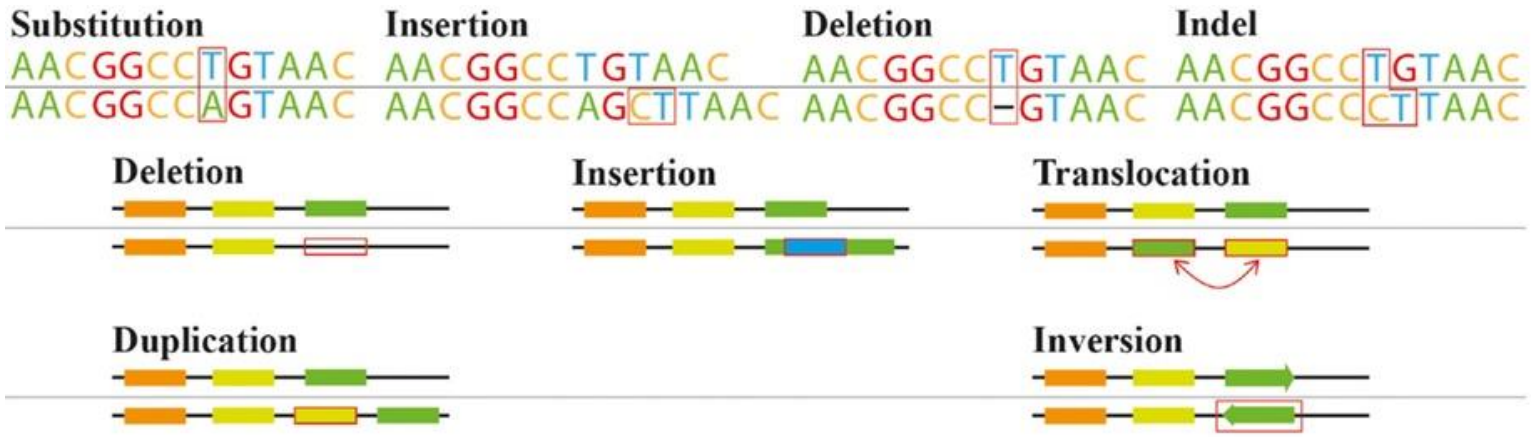
Construct genome sequence from overlaps between reads

What does resequencing give you?

- Alignment profile of your sequence on the genome reference
- Small genomic variant profile. (SNP/Indel)
- Large genomic structural variant profile.
- Annotation and functional effect prediction of variant.

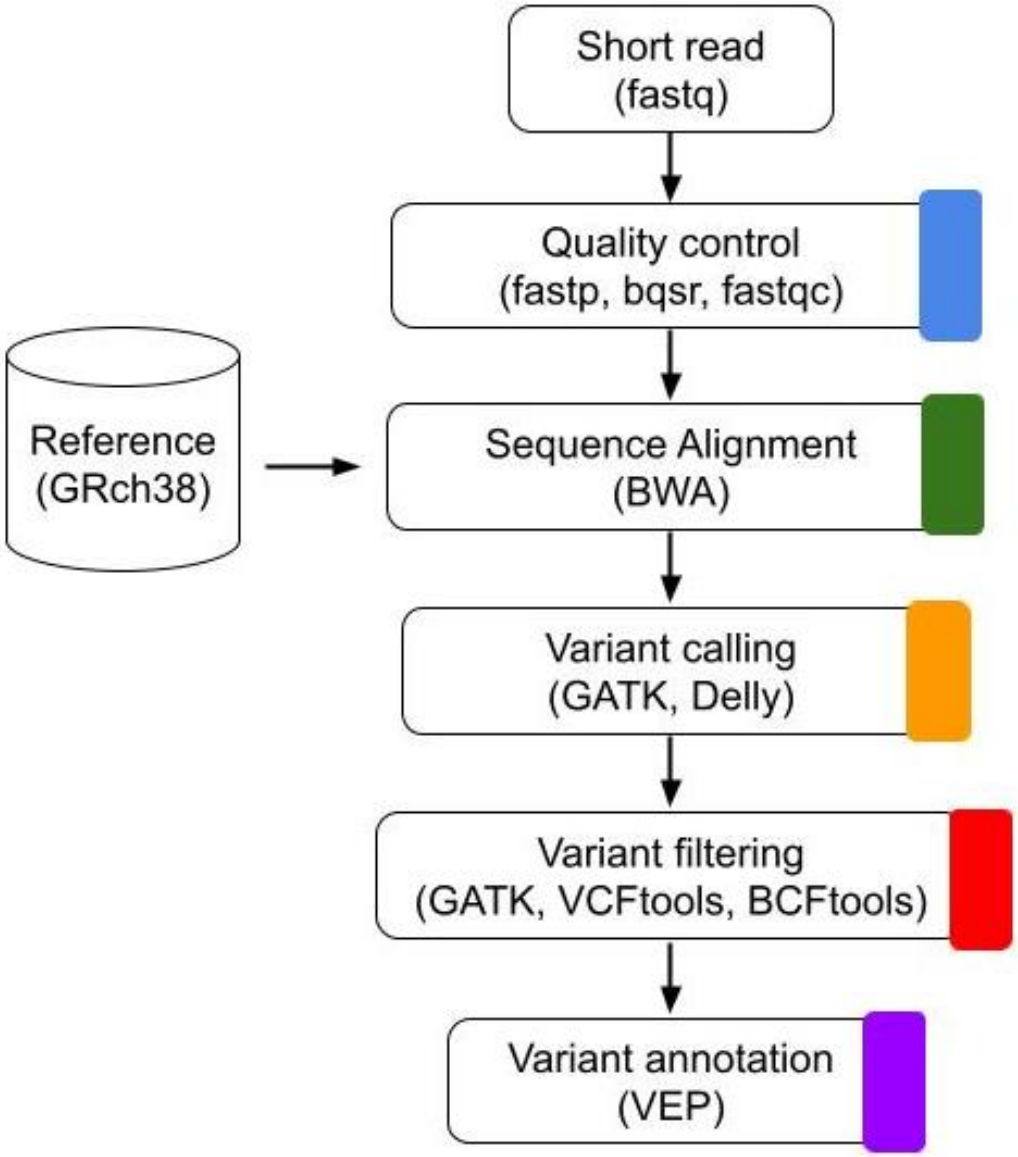


https://www.ensembl.org/info/genome/variation/prediction/consequences.svg



Cardoso JGR, Andersen MR, Herrgård MJ and Sonnenschein N (2015) Analysis of genetic variation and potential applications in genome-scale metabolic modeling. *Front. Bioeng. Biotechnol.* **3**:13. doi: 10.3389/fbioe.2015.00013

Bioinformatics workflow



Sequence file format

FASTA

>title
seqeunce

```
>Chromosome dna_sm:chromosome chromosome:ASM19595v2:Chromosome:1:4411532:1 REF
TTGACCGATGACCCCGGTTCAGGCTTCACCACAGTGTGGAACGCGGTCGTCTCCGAACTT
```

FASTQ

@title
sequence
+optional_text
quality

```
@SRR11662136.1 1 length=150
NCTTGATCGACTTGTCAGAATTCCGGCGACTGGCGCGGTTTCGGCGCAGCGAACTCAT
+SRR11662136.1 1 length=150
#AAAFFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJFJJFJJJJJJJJJJ
```

Phred quality scores

```
@SRR11662136.1 1 length=150
NCTTGATCGACTTGTCAGAATTCCGGCGACTGGCGCGGTTTCGGCGCAGCGAACTCAT
+SRR11662136.1 1 length=150
#AAAFFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJFJJFJJJJJJJJJJ
```

ASCII_BASE=33 Illumina, Ion Torrent, PacBio and Sanger

Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII
0	1.00000	33 !	11	0.07943	44 ,	22	0.00631	55 7	33	0.00050	66 B
1	0.79433	34 "	12	0.06310	45 -	23	0.00501	56 8	34	0.00040	67 C
2	0.63096	35 #	13	0.05012	46 .	24	0.00398	57 9	35	0.00032	68 D
3	0.50119	36 $	14	0.03981	47 /	25	0.00316	58 :	36	0.00025	69 E
4	0.39811	37 %	15	0.03162	48 0	26	0.00251	59 ;	37	0.00020	70 F
5	0.31623	38 &	16	0.02512	49 1	27	0.00200	60 <	38	0.00016	71 G
6	0.25119	39 '	17	0.01995	50 2	28	0.00158	61 =	39	0.00013	72 H
7	0.19953	40 (	18	0.01585	51 3	29	0.00126	62 >	40	0.00010	73 I
8	0.15849	41)	19	0.01259	52 4	30	0.00100	63 ?	41	0.00008	74 J
9	0.12589	42 *	20	0.01000	53 5	31	0.00079	64 @	42	0.00006	75 K
10	0.10000	43 +	21	0.00794	54 6	32	0.00063	65 A			

ASCII_BASE=64 Old Illumina

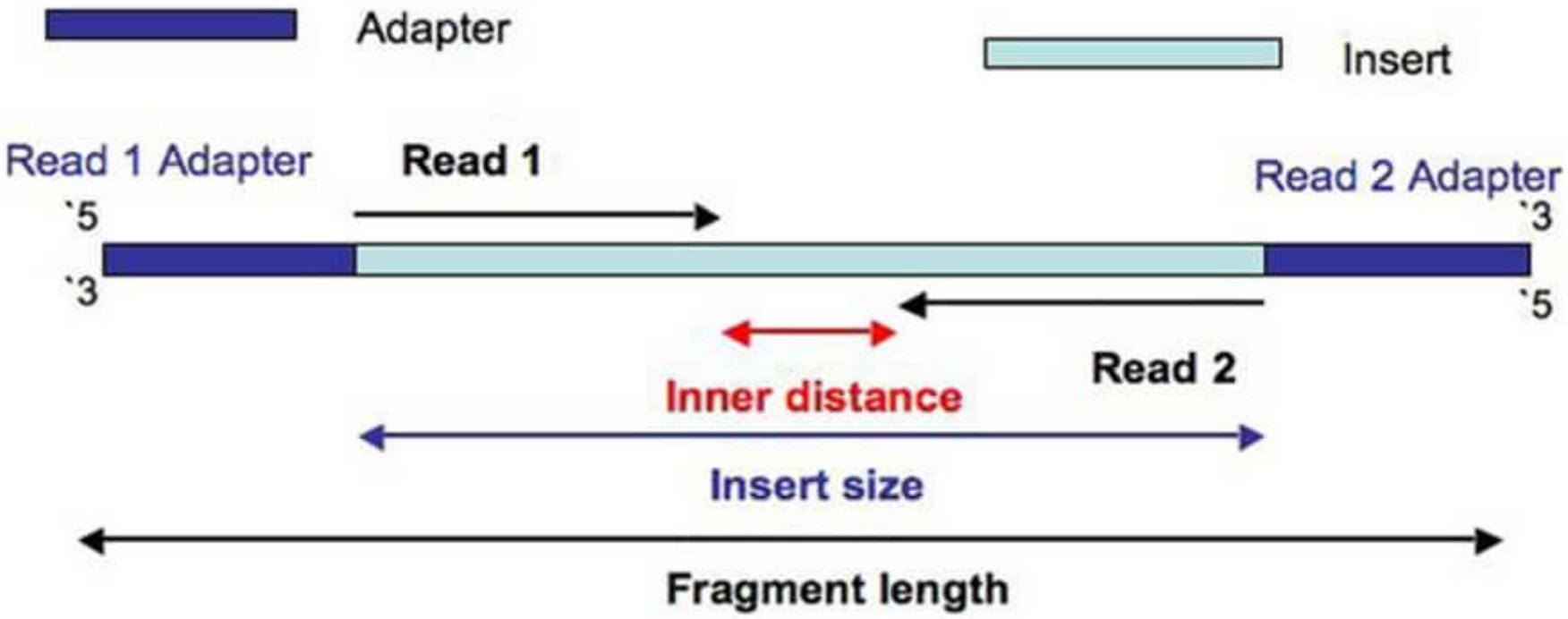
Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII	Q	P_error	ASCII
0	1.00000	64 @	11	0.07943	75 K	22	0.00631	86 V	33	0.00050	97 a
1	0.79433	65 A	12	0.06310	76 L	23	0.00501	87 W	34	0.00040	98 b
2	0.63096	66 B	13	0.05012	77 M	24	0.00398	88 X	35	0.00032	99 c
3	0.50119	67 C	14	0.03981	78 N	25	0.00316	89 Y	36	0.00025	100 d
4	0.39811	68 D	15	0.03162	79 O	26	0.00251	90 Z	37	0.00020	101 e
5	0.31623	69 E	16	0.02512	80 P	27	0.00200	91 [	38	0.00016	102 f
6	0.25119	70 F	17	0.01995	81 Q	28	0.00158	92 \	39	0.00013	103 g
7	0.19953	71 G	18	0.01585	82 R	29	0.00126	93]	40	0.00010	104 h
8	0.15849	72 H	19	0.01259	83 S	30	0.00100	94 ^	41	0.00008	105 i
9	0.12589	73 I	20	0.01000	84 T	31	0.00079	95 _	42	0.00006	106 j
10	0.10000	74 J	21	0.00794	85 U	32	0.00063	96 `			

Phred Q = $10\log_{10} P$ ($P = 10^{-Q/10}$)

e.g. **Q30** $P = 10^{-3}$ or ***1 in 1000 bp***

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%
60	1 in 1,000,000	99.9999%

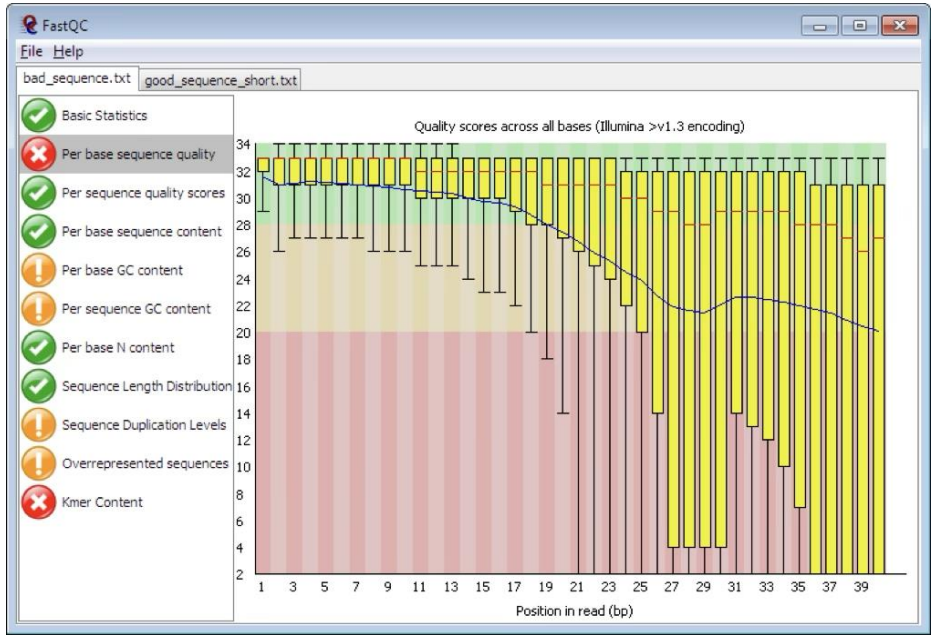
Short read seqeunce



Evaluate sequence quality (Short read)

FastQC

https://github.com/s-andrews/FastQC



Good sequencequality

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/good_sequence_short_fastqc.html

Bad sequence quality

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/bad_sequence_fastqc.html

Evaluate sequence quality (Short read)

Things to focus

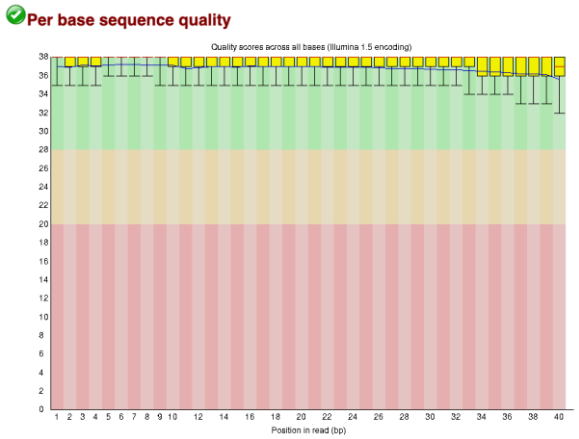
- Number of reads
- Number of bases (Can use to calculate sequence coverage)
  (Total bases / Genome size) = coverage
- Sequence length

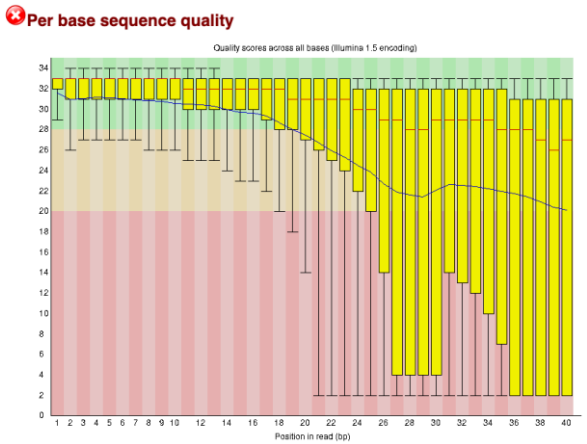
Basic Statistics

Measure	Value
Filename	good_sequence_short.txt
File type	Conventional base calls
Encoding	Illumina 1.5
Total Sequences	250000
Total Bases	10 Mbp
Sequences flagged as poor quality	0
Sequence length	40
%GC	45

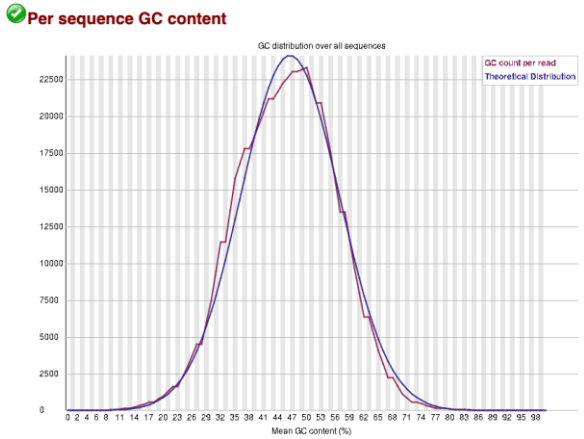
Evaluate sequence quality (Short read)

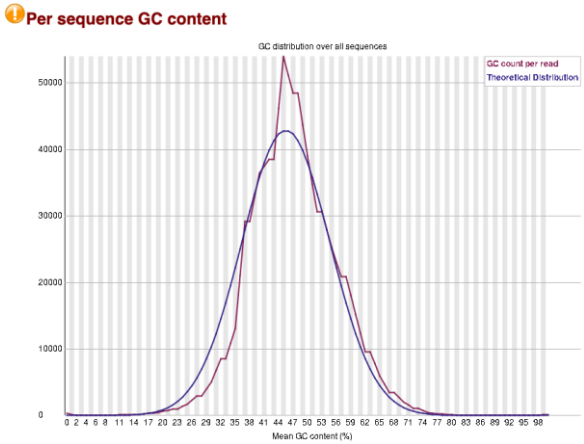
- Base sequence quality





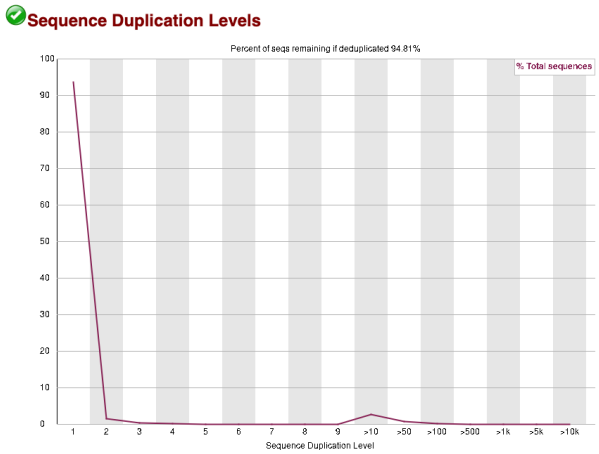
- GC content

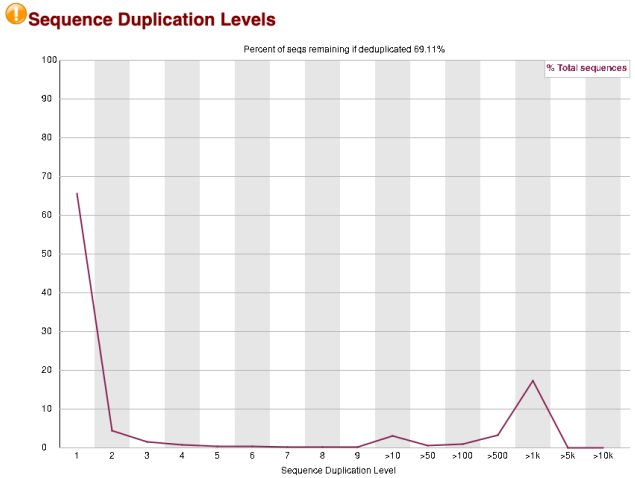




Evaluate sequence quality (Short read)

- Seuqence Duplication





- Overrepresented sequences

Overrepresented sequences

No overrepresented sequences

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
AGAGTTTTATCGCTTCCATGACGCAGAAGTTAACACTTTC	2065	0.5224039181558763	No Hit
GATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCATG	2047	0.5178502762542754	No Hit
ATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCATGA	2014	0.5095019327680071	No Hit
CGATAAAAATGATTGGCGTATCCAACCTGCAGAGTTTTAT	1913	0.4839509420979134	No Hit
GTATCCAACCTGCAGAGTTTTATCGCTTCCATGACGCAGA	1879	0.4753496185060006 6	No Hit
AAAAATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCT	1846	0.4670012750197325	No Hit
TGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCAT	1841	0.4657363744915099 5	No Hit
AACCTGCAGAGTTTTATCGCTTCCATGACGCAGAAGTTAA	1836	0.4644714739632875 3	No Hit
GATAAAAATGATTGGCGTATCCAACCTGCAGAGTTTTATC	1831	0.4632065734350651	No Hit
ATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCCA	1779	0.4500516079415514 7	No Hit
AAATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTC	1779	0.4500516079415514 7	No Hit
AATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTTCC	1760	0.4452449859343061	No Hit
AAAATGATTGGCGTATCCAACCTGCAGAGTTTTATCGCTT	1729	0.4374026026593269	No Hit
CGTATCCAACCTGCAGAGTTTTATCGCTTCCATGACGCAG	1713	0.4333549209690149 6	No Hit
ATCCAACCTGCAGAGTTTTATCGCTTCCATGACGCAGAAG	1708	0.4320908204407925 3	No Hit
CAGAGTTTTATCGCTTCCATGACGCAGAAGTTAACACTTT	1684	0.4260184979053247 6	No Hit
TGCAGAGTTTTATCGCTTCCATGACGCAGAAGTTAACACT	1668	0.421970816215012 8	No Hit
CAACCTGCAGAGTTTTATCGCTTCCATGACGCAGAAGTTA	1668	0.421970816215012 8	No Hit
TATCCAACCTGCAGAGTTTTATCGCTTCCATGACGCAGAA	1630	0.412357572200522 1	No Hit
GTCATGGAAGCGATAAAACTCTGCAGGTTGGATACGCCAA	1620	0.4098277711440772 6	No Hit

Evaluate sequence quality (Short read)

```
fastqc -t ${task.cpus} \
       -f fastq \
       -o . \
       ${fastqs}
```

-t : Number of cpu

-f : Input file format (BAM, SAM, and fastq)

-o : Output directory

${fastq} = Input fastq file

Quality filtering

fastp

https://github.com/OpenGene/fastp

A tool designed to provide ultrafast all-in-one preprocessing and quality control for FastQ data. Suitable for short read sequencing.

```
fastp -i file_1.fq.gz \
    -I file_2.fq.gz \
    -q 30 \
    -o clean_file_1.fq.gz \
    -O clean_file_2.fq.gz \
    -h fastp_report.html
```

-i : Input read_1

-I : Input read_2

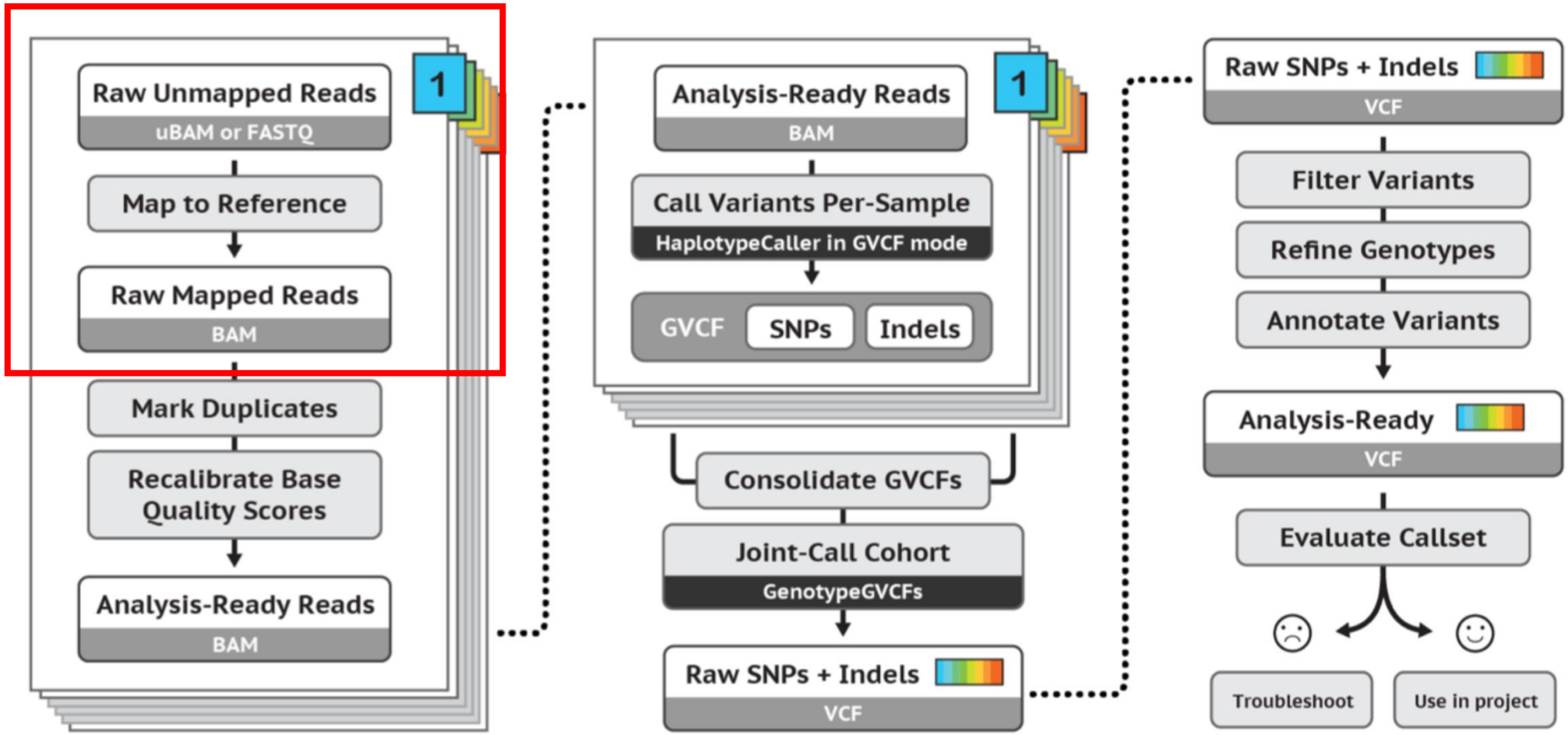
-q : Quality value that based is qualified

-o : Output clean read_1

-O : Output clean read_2

-h : fastp report in html format

GATK Best Practice



https://gatk.broadinstitute.org/hc/en-us/articles/360035535932-Germline-short-variant-discovery-SNPs-Indels-

Seqeunce Alignment

- Burrows-Wheelers Alignment (BWA)

How does it work?

Example of Burrow-Wheelers Transform

AATCGT
(Reference)

All Rotation

0	AATCGT$
1	ATCGT$A
2	TCGT$AA
3	CGT$AAT
4	GT$AATC
5	T$AATCG
6	$AATCGT

Sort

6	$AATCGT
0	AATCGT$
1	ATCGT$A
3	CGT$AAT
4	GT$AATC
5	T$AATCG
2	TCGT$AA

Suffix Array

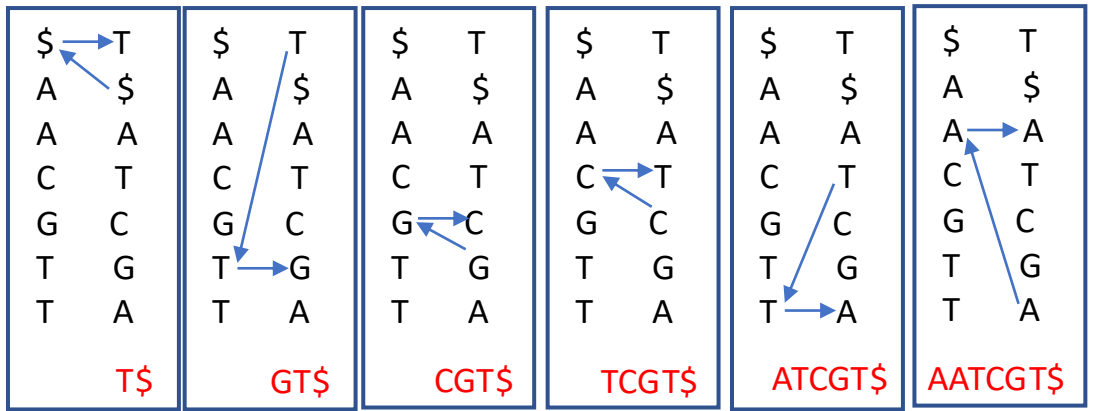
BWT → T$ATCGA

6	$AATCGT
0	AATCGT$
1	ATCGT$A
3	CGT$AAT
4	GT$AATC
5	T$AATCG
2	TCGT$AA

Example inverse tranformation

T$ATCGA

For any x character, i^{th} x character in first column.
Correspond to i^{th} x character in last column.



AATCGT
(Original sequence)

https://medium.com/@mr-easy/burrows-wheeler-alignment-part-1-eb93057bfff5

Seqeunce Alignment

Burrows-Wheelers Alignment (BWA)

How does it work?

- Mapping process start from the last character of query sequence.
- When band size reduce to zero, it mean no match for that x character.
- Calculate next band : New start = $C(x) + Rank(x, start)$ and New stop = $C(x) + Rank(x, stop)$
- $C(x)$: How many characters in the entire seqeunce that are alphabetically smaller than x.
- $Rank(x, i)$: How many times character x present before index i.

BWT data structure

6	$AATCGT
0	AATCGT$
1	ATCGT$A
3	CGT$AAT
4	GT$AATC
5	T$AATCG
2	TCGT$AA

Suffix Array

Query : ATC

6	0	$ . . . . . T
0	1	A . . . . . $
1	2	A . . . . . A
3	3	C . . . . . T
4	4	G C
5	5	T . . . . . G
2	6	T . . . . . A

First band = [3,4)

$Rank(T, 3) = 1$
$Rank(T, 4) = 2$

Query : ATC

0	$ . . . . . T
1	A . . . . . $
2	A . . . . . A
3	C . . . . . T
4	G C
5	T . . . . . G
6	TC . . . . A

New band = [5+1,5+2) = [6,7)

$Rank(A, 6) = 1$
$Rank(A, 7) = 2$

Query : ATC

0	$ . . . . . T
1	A . . . . . $
2	ATC . . . A
3	C . . . . . T
4	G C
5	T . . . . . G
6	TC . . . . A

New band = [1+1,1+2) = [2,3)

Look back at suffix Array.
Our query map at index 1.

A	**A**	T	C	G	T
0	**1**	2	3	4	5

https://medium.com/@mr-easy/burrows-wheeler-alignment-part-1-eb93057bfff5

Seqeunce Alignment

Burrows-Wheelers Alignment (BWA)

How does it work?

- Mapping process start from the last character of query sequence.
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- $C(x)$: How many characters in the entire seqeunce that are alphabetically smaller than x.
- $Rank(x, i)$: How many times character x present before index i.

BWT data structure

6	$AATCGT
0	AATCGT$
1	ATCGT$A
3	CGT$AAT
4	GT$AATC
5	T$AATCG
2	TCGT$AA

Suffix Array

Query : ACG

6	0	$ . . . . . T
0	1	A . . . . . $
1	2	A . . . . . A
3	3	C . . . . . T
4	4	G C
5	5	T . . . . . G
2	6	T . . . . . A

First band = [4,5)

$Rank(C, 4)$ = 0
$Rank(C, 5)$ = 1

Query : ACG

0	$ . . . . . T
1	A . . . . . $
2	A . . . . . A
3	CG . . . . T
4	G C
5	T . . . . . G
6	T . . . . . A

New band = [3+0,3+1) = [3,4)

$Rank(A, 3)$ = 1
$Rank(A, 4)$ = 1

Query : ACG

0	$ . . . . . T
1	A . . . . . $
2	A . . . . . A
3	C . . . . . T
4	G C
5	T . . . . . G
6	T . . . . . A

New band = [1+1,1+1) = [2,2)

Band size = 0 ; No match.

A	A	T	**C**	G	T
0	1	2	**3**	4	5

https://medium.com/@mr-easy/burrows-wheeler-alignment-part-1-eb93057bfff5

Alignment file format (SAM/BAM)

Header

Alignment section

```
@SQ     SN:Chromosome   LN:4411532
@HD     VN:1.5  SO:unsorted     GO:query
@RG     ID:ERR846997    PL:ILLUMINA     LB:lib1 SM:ERR846997
@PG     ID:bwa  PN:bwa  VN:0.7.18-r1243-dirty   CL:bwa mem -M -t 6 -R @RG\tID:ERR846997\tPL:ILLUMINA\tLB:lib1\tSM:ERR846997 -o ./analysis/ERR846997
.sam ../reference/Mycobacterium_tuberculosis_h37rv.ASM19595v2.dna_sm.toplevel.fa ../data/ERR846997_1.fastq.gz ../data/ERR846997_2.fastq.gz
ERR846997.1     99      Chromosome      20      60      100M    =       72      152     CAGGCTTCACCACAGTGTGGAACGCGGTCGTCTCCGAACTTAACGGCGACCCTAAGGTT
GACGACGGACCCAGCAGTGATGCTAATCTCAGCGCTCCGCT       @C@FDFFFGHGHHHIGIIIJJJJJJIGIIGIHIJJIGGIIJIIDEHHEDDDDDD5>CCCCCDDDDDDBDDBDDBDDCCDDDDCCDDCDDCDCDDDDD@@
<       NM:i:0  MD:Z:100        MC:Z:100M       MQ:i:60 AS:i:100        XS:i:0  RG:Z:ERR846997
ERR846997.1     147     Chromosome      72      60      100M    =       20      -152    TAAGGTTGACGACGGACCCAGCAGTGATGCTAATCTCAGCGCTCCGCTGACCCCTCAGC
AAAGGGCTTGGCTCAATCTCGTCCAGCCATTGACCATCGTC       CDDDDDDB<@A<B?<A@DC@CCDCC:DC?DCCAA92=<?;9AEFEEEGA;HGG=GGHEBIIIGHIEHDGGHCGGBGGGIHDDHIIGEHHHFDFFDFF@@
@       NM:i:0  MD:Z:100        MC:Z:100M       MQ:i:60 AS:i:100        XS:i:0  RG:Z:ERR846997
ERR846997.2     83      Chromosome      97      60      100M    =       48      -149    GATGCTAAGCTCAGCGCTCCGGTGTCCCCTCAGCAAAGGGCTTGGCTCAATCTCGTCCA
GCCATTGACCATCGTCGAGGGGTTTGCTCTGTTATCCGTGC       (4AAAA>4(2&9(0959;0)+(20-5855(::5:;>?>?@>@>@?7)7754->A@=AC>@CB8?DBD08?86?:)?8FFA;EEEA+@<:<?<0)@DB?;
=       NM:i:3  MD:Z:8T12C2A75  MC:Z:100M       MQ:i:60 AS:i:85 XS:i:0  RG:Z:ERR846997
ERR846997.2     163     Chromosome      48      60      100M    =       97      149     CGTCTCCGAACTTAACGGCGACCCTAAGGTTGACGACGGACCCAGCAGTGATGCTAATC
TCAGCGCTCCGCTGACCCCTCAGCAAAGGGCTTGGCTCAAT       ?@:BD?:@12<2AF@?:CE:??00?DFFFB4B=)5;BB8<?BC?=@B=(:@BA@:A:ABB5>>B@0?>>5;;&0<A?BBB8>ABBB>9?22<)+>>@>A
B       NM:i:0  MD:Z:100        MC:Z:100M       MQ:i:60 AS:i:100        XS:i:0  RG:Z:ERR846997
```

Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	`[!-?A-~]{1,254}`	Query template NAME
2	FLAG	Int	$[0,2^{16}-1]$	bitwise FLAG
3	RNAME	String	`\*|[!-()+-<>-~][!-~]*`	Reference sequence NAME
4	POS	Int	$[0,2^{31}-1]$	1-based leftmost mapping POSition
5	MAPQ	Int	$[0,2^{8}-1]$	MAPping Quality
6	CIGAR	String	`\*|([0-9]+[MIDNSHPX=])+`	CIGAR string
7	RNEXT	String	`\*|=|[!-()+-<>-~][!-~]*`	Ref. name of the mate/next read
8	PNEXT	Int	$[0,2^{31}-1]$	Position of the mate/next read
9	TLEN	Int	$[-2^{31}+1,2^{31}-1]$	observed Template LENgth
10	SEQ	String	`\*|[A-Za-z=.]+`	segment SEQuence
11	QUAL	String	`[!-~]+`	ASCII of Phred-scaled base QUALity+33

http://samtools.github.io/hts-specs/SAMv1.pdf

Alignment file format (SAM/BAM)

Column 2 : Flag

Bitwise Flag

Bit		Description
1	0x1	template having multiple segments in sequencing
2	0x2	each segment properly aligned according to the aligner
4	0x4	segment unmapped
8	0x8	next segment in the template unmapped
16	0x10	SEQ being reverse complemented
32	0x20	SEQ of the next segment in the template being reverse complemented
64	0x40	the first segment in the template
128	0x80	the last segment in the template
256	0x100	secondary alignment
512	0x200	not passing filters, such as platform/vendor quality controls
1024	0x400	PCR or optical duplicate
2048	0x800	supplementary alignment

READ2 PROPER_PAIR

163 => 0000 1010 0011

PAIRED

MREVERSE

Column 6 : CIGAR

Op	BAM	Description	Consumes query	Consumes reference
M	0	alignment match (can be a sequence match or mismatch)	yes	yes
I	1	insertion to the reference	yes	no
D	2	deletion from the reference	no	yes
N	3	skipped region from the reference	no	yes
S	4	soft clipping (clipped sequences present in SEQ)	yes	no
H	5	hard clipping (clipped sequences NOT present in SEQ)	no	no
P	6	padding (silent deletion from padded reference)	no	no
=	7	sequence match	yes	yes
X	8	sequence mismatch	yes	yes

100M2D1I10M5S

100 maped | 2 delete | 1 insert | 10 maped | 5 soft clip

http://samtools.github.io/hts-specs/SAMv1.pdf

Sequence Alignment

Burrows-Wheelers Aligner (BWA)

```
Program: bwa (alignment via Burrows-Wheeler transformation)
Version: 0.7.17-r1188
Contact: Heng Li <lh3@sanger.ac.uk>

Usage:   bwa <command> [options]

Command: index         index sequences in the FASTA format
         mem           BWA-MEM algorithm
         fastmap       identify super-maximal exact matches
         pemerge       merge overlapping paired ends (EXPERIMENTAL)
         aln           gapped/ungapped alignment
         samse         generate alignment (single ended)
         sampe         generate alignment (paired ended)
         bwasw         BWA-SW for long queries

         shm           manage indices in shared memory
         fa2pac        convert FASTA to PAC format
         pac2bwt       generate BWT from PAC
         pac2bwtgen    alternative algorithm for generating BWT
         bwtupdate     update .bwt to the new format
         bwt2sa        generate SA from BWT and Occ

Note: To use BWA, you need to first index the genome with `bwa index'.
      There are three alignment algorithms in BWA: `mem', `bwasw', and
      `aln/samse/sampe'. If you are not sure which to use, try `bwa mem'
      first. Please `man ./bwa.1' for the manual.
```

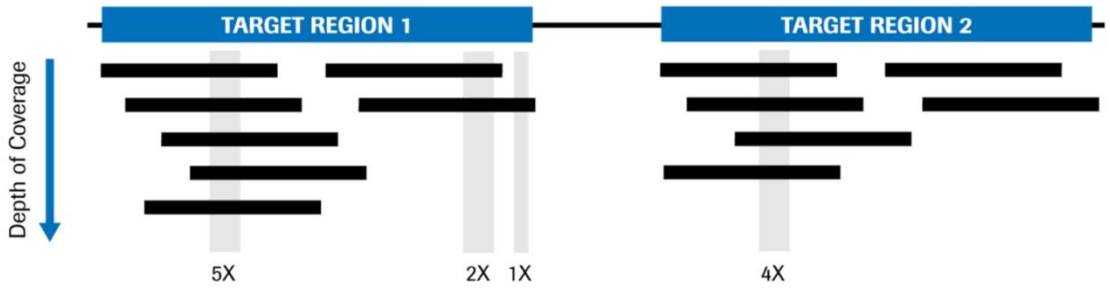
Create bwa index of reference sequence

Maximally Exact Matches (mem) algorithm

https://github.com/lh3/bwa

Li H. (2013) Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. arXiv:1303.3997v2 [q-bio.GN]

Sequence Alignment



Index reference seqeunce

```
bwa index ${reference_fasta}
```

- Input : sequence file in fasta format
- Output : bwa index file in format (.amb, .ann, .bwt, .pac, .sa)

Align input seqeunce with reference sequence

```
bwa mem -M -t ${threads} \
        -R "@RG\tID:${id}\tPL:ILLUMINA\tLB:${lb}\tSM:${sm}" \
        -o ${out_sam} \
        ${reference} \
        ${read1} \
        ${read2}
```

-R : Define @RG header in SAM file.

-M : Mark shorter split hits as secondary

-o : Output SAM file

-t : Number of threads

https://sequencing.roche.com/us/en/article-listing/metrics-for-target-enriched-ngs.html

Alignment Post-processing

SAM/BAM file manipulation with Samtools

```
Program: samtools (Tools for alignments in the SAM format)
Version: 1.7 (using htslib 1.7)

Usage:   samtools <command> [options]

Commands:
  -- Indexing
     dict           create a sequence dictionary file
     faidx          index/extract FASTA
     index          index alignment

  -- Editing
     calmd          recalculate MD/NM tags and '=' bases
     fixmate        fix mate information
     reheader       replace BAM header
     targetcut      cut fosmid regions (for fosmid pool only)
     addreplacerg   adds or replaces RG tags
     markdup        mark duplicates

  -- File operations
     collate        shuffle and group alignments by name
     cat            concatenate BAMs
     merge          merge sorted alignments
     mpileup        multi-way pileup
     sort           sort alignment file
     split          splits a file by read group
     quickcheck     quickly check if SAM/BAM/CRAM file appears intact
     fastq          converts a BAM to a FASTQ
     fasta          converts a BAM to a FASTA
```

```
  -- Statistics
     bedcov         read depth per BED region
     depth          compute the depth
     flagstat       simple stats
     idxstats       BAM index stats
     phase          phase heterozygotes
     stats          generate stats (former bamcheck)

  -- Viewing
     flags          explain BAM flags
     tview          text alignment viewer
     view           SAM<->BAM<->CRAM conversion
     depad          convert padded BAM to unpadded BAM
```

Alignment Post-processing

Get alignment stats of SAM/BAM file.

```
samtools flagstat -@ ${threads} ${sam_file}
```

-@ : number of threads

Example flagstats output

```
1031152 + 0 in total (QC-passed reads + QC-failed reads)
188 + 0 secondary
0 + 0 supplementary
0 + 0 duplicates
1030944 + 0 mapped (99.98% : N/A)
1030964 + 0 paired in sequencing
515482 + 0 read1
515482 + 0 read2
1029364 + 0 properly paired (99.84% : N/A)
1030548 + 0 with itself and mate mapped
208 + 0 singletons (0.02% : N/A)
0 + 0 with mate mapped to a different chr
0 + 0 with mate mapped to a different chr (mapQ>=5)
```

Convert SAM to BAM.

```
samtools view -bS -@ ${threads} -o ${bam_file} ${sam_file}
```

-b : output as BAM format.
-S : input format is auto-detected.
-@ : number of threads.
-o : output BAM file.

Sort BAM file (by position).

```
samtools sort -@ ${threads} -o ${sort_bam_file} ${bam_file}
```

-@ : number of threads
-o : output filename

Alignment Post-processing

Index BAM file.

```
samtools index -@ ${threads} ${sort_bam_file}
```

-@ : number of threads

Filter unmapped reads.

```
samtools view -b -F 4 -F 8 -@ ${threads} -o ${output_bam_file} ${input_bam_file}
```

-b : output as BAM format.
-F : include reads with none of the FLAGS in INT present
-@ : number of threads.
-o : output filename

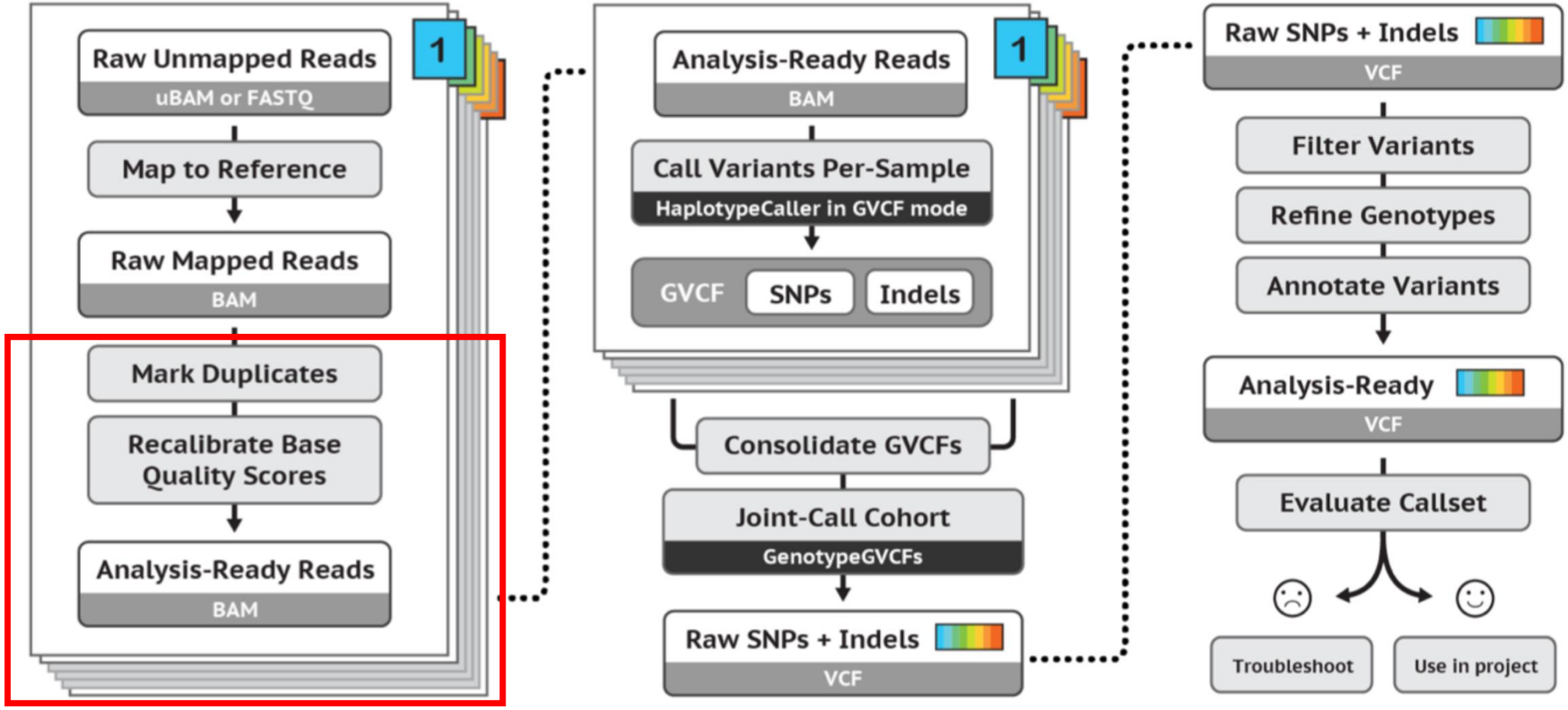
Bit		Description
1	0x1	template having multiple segments in sequencing
2	0x2	each segment properly aligned according to the aligner
4	0x4	segment unmapped
8	0x8	next segment in the template unmapped
16	0x10	SEQ being reverse complemented
32	0x20	SEQ of the next segment in the template being reverse complemented
64	0x40	the first segment in the template
128	0x80	the last segment in the template
256	0x100	secondary alignment
512	0x200	not passing filters, such as platform/vendor quality controls
1024	0x400	PCR or optical duplicate
2048	0x800	supplementary alignment

Explain BAM flags.

```
samtools flags ${flags_number}
```

Example output

```
(conda_py3) admin@7a82b97995e6:~$ samtools flags 163
0xa3     163     PAIRED,PROPER_PAIR,MREVERSE,READ2
```

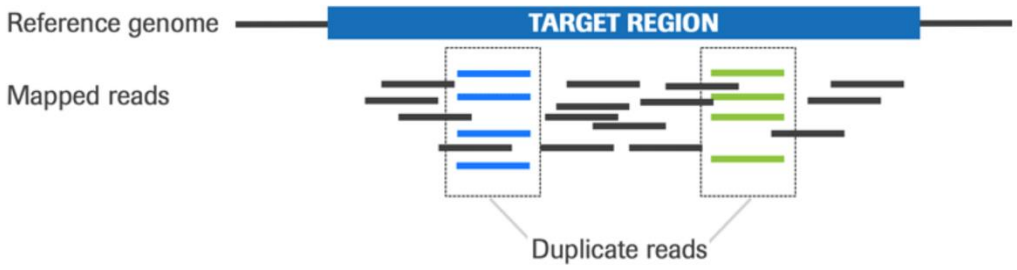

Raw Unmapped Reads
uBAM or FASTQ
1
Map to Reference
Raw Mapped Reads
BAM
Mark Duplicates
Recalibrate Base Quality Scores
Analysis-Ready Reads
BAM
Analysis-Ready Reads
BAM
1
Call Variants Per-Sample
HaplotypeCaller in GVCF mode
GVCF
SNPs
Indels
Consolidate GVCFs
Joint-Call Cohort
GenotypeGVCFs
Raw SNPs + Indels
VCF
Raw SNPs + Indels
VCF
Filter Variants
Refine Genotypes
Annotate Variants
Analysis-Ready
VCF
Evaluate Callset
Troubleshoot
Use in project

Alignment Post-processing

Markduplicate with PICARD

- To locate and tags duplicate reads in BAM or SAM file.
- Each duplication set of reads will be consider as one.

```
gatk --java-options "-Xmx4G" MarkDuplicates \
        -I ${sort_bam_file} \
        -O ${marked_dup_bam_file} \
        -M ${output_metrics_file} \
        --TMP_DIR ${tmp_dir}
```



--java-options : Define specific java option. In this case -Xmx mean maximum memory.
-I : input coordinated sorted BAM file.
-O : output BAM file.
-M : output metric file.
--TMP_DIR : specify space available directory for temprary storage of working file.

https://sequencing.roche.com/us/en/article-listing/metrics-for-target-enriched-ngs.html

Alignment Post-processing

Markduplicate with PICARD

Before

```
HWI-ST1217:277:D2D19ACXX:2:1216:9551:51089	163	chr14	18967702	4	101M	=	18967757	1
56	AGGGGGAGCGGCAAGAGAAACGCGGGCACTTCTGGAGACCACGAAGAATCTGCTATGAAGACACTCAGGAGCAAGATGGGAAAGAGGTGCTGCCACTGCTC	=;;;=A8A
AAA88A;?:)==>=(<5=<<@@@##########################################################	NM:i:7	MD:Z:17C4T21C2C3
2C3T15T0	MC:Z:101M	AS:i:70 XS:i:65 RG:Z:A	XA:Z:chr14,+18967591,101M,8;chr14,-19434084,101M,8;chr14,-194339
73,101M,8;
```

After

```
HWI-ST1217:277:D2D19ACXX:2:1216:9551:51089	1187	chr14	18967702	4	101M	=	18967757	1
56	AGGGGGAGCGGCAAGAGAAACGCGGGCACTTCTGGAGACCACGAAGAATCTGCTATGAAGACACTCAGGAGCAAGATGGGAAAGAGGTGCTGCCACTGCTC	=;;;=A8A
AAA88A;?:)==>=(<5=<<@@@##########################################################	XA:Z:chr14,+18967591,101
M,8;chr14,-19434084,101M,8;chr14,-19433973,101M,8;	MC:Z:101M	MD:Z:17C4T21C2C32C3T15T0	PG:Z:MarkDuplica
tes	RG:Z:A	NM:i:7	AS:i:70 XS:i:65
```

BAM flags

```
(conda_py3) admin@7a82b97995e6:~$ samtools flags 1187
0x4a3	1187	PAIRED,PROPER_PAIR,MREVERSE,READ2,DUP
```

Bit		Description
1	0x1	template having multiple segments in sequencing
2	0x2	each segment properly aligned according to the aligner
4	0x4	segment unmapped
8	0x8	next segment in the template unmapped
16	0x10	SEQ being reverse complemented
32	0x20	SEQ of the next segment in the template being reverse complemented
64	0x40	the first segment in the template
128	0x80	the last segment in the template
256	0x100	secondary alignment
512	0x200	not passing filters, such as platform/vendor quality controls
1024	0x400	PCR or optical duplicate
2048	0x800	supplementary alignment

Alignment Post-processing

Collect coverage and performance metrics of BAM file with CollectWgsMetrics

```
gatk --java-options "-Xmx4G" CollectWgsMetrics \
        -MQ 20 -Q 20 \
        -R ${reference_fasta} \
        -O ${output_metrics_file} \
        -I ${BAM_file}
```

--java-options : Define specific java option.
-MQ : Minimum mapping quality.
-Q : Minimum base quality.
-R : reference fasta file.
-O : plain text output file.
-I : input BAM file.

Metric for evaluating the performance of whole genome sequening experiments.

Field	Description
GENOME_TERRITORY	The number of non-N bases in the genome reference over which coverage will be evaluated.
MEAN_COVERAGE	The mean coverage in bases of the genome territory, after all filters are applied.
SD_COVERAGE	The standard deviation of coverage of the genome after all filters are applied.
MEDIAN_COVERAGE	The median coverage in bases of the genome territory, after all filters are applied.
MAD_COVERAGE	The median absolute deviation of coverage of the genome after all filters are applied.
PCT_EXC_MAPQ	The fraction of aligned bases that were filtered out because they were in reads with low mapping quality (default is < 20).
PCT_EXC_DUPE	The fraction of aligned bases that were filtered out because they were in reads marked as duplicates.
PCT_EXC_UNPAIRED	The fraction of aligned bases that were filtered out because they were in reads without a mapped mate pair.
PCT_EXC_BASEQ	The fraction of aligned bases that were filtered out because they were of low base quality (default is < 20).
PCT_EXC_OVERLAP	The fraction of aligned bases that were filtered out because they were the second observation from an insert with overlapping reads.
PCT_EXC_CAPPED	The fraction of aligned bases that were filtered out because they would have raised coverage above the capped value (default cap = 250x).
PCT_EXC_TOTAL	The total fraction of aligned bases excluded due to all filters.
PCT_1X	The fraction of bases that attained at least 1X sequence coverage in post-filtering bases.
PCT_5X	The fraction of bases that attained at least 5X sequence coverage in post-filtering bases.
PCT_10X	The fraction of bases that attained at least 10X sequence coverage in post-filtering bases.
PCT_15X	The fraction of bases that attained at least 15X sequence coverage in post-filtering bases.
PCT_20X	The fraction of bases that attained at least 20X sequence coverage in post-filtering bases.
PCT_25X	The fraction of bases that attained at least 25X sequence coverage in post-filtering bases.
PCT_30X	The fraction of bases that attained at least 30X sequence coverage in post-filtering bases.
PCT_40X	The fraction of bases that attained at least 40X sequence coverage in post-filtering bases.
PCT_50X	The fraction of bases that attained at least 50X sequence coverage in post-filtering bases.
PCT_60X	The fraction of bases that attained at least 60X sequence coverage in post-filtering bases.
PCT_70X	The fraction of bases that attained at least 70X sequence coverage in post-filtering bases.
PCT_80X	The fraction of bases that attained at least 80X sequence coverage in post-filtering bases.
PCT_90X	The fraction of bases that attained at least 90X sequence coverage in post-filtering bases.
PCT_100X	The fraction of bases that attained at least 100X sequence coverage in post-filtering bases.
HET_SNP_SENSITIVITY	The theoretical HET SNP sensitivity.
HET_SNP_Q	The Phred Scaled Q Score of the theoretical HET SNP sensitivity.

Alignment Post-processing

Base Quality Score Recalibration (BQSR)

- There are over- or under-estimated base quality scores in the data.
- Build model of covariation based on the input data and a set of known variants.
- Adjust the base quality scores in the data based on the model.
- Known variant resource : https://console.cloud.google.com/storage/browser/gcp-public-data--broad-references

Addition preparation on reference fasta

- Create index of reference fasta file (.fai).

```
samtools faidx ${reference_fasta_file}
```

- Create a sequence dictionary for reference sequence (.dict).

```
gatk --java-options "-Xmx4G" CreateSequenceDictionary -R ${reference_fasta_file}
```

Alignment Post-processing

Base Quality Score Recalibration (BQSR)

```
gatk --java-options "-Xmx4G" BaseRecalibrator \
        -R ${reference_fasta} \
        -I ${BAM_file} \
        -O ${BQSR_report} \
        --known-sites ${known_sites_vcf} \
        --known-sites ${known_sites_vcf}
```

-R : reference fasta file.
-I : input BAM file.
-O : ouptut BQSR report file.
--known-sites : know variant VCF file.

```
gatk --java-options "-Xmx4G" ApplyBQSR \
        -I ${BAM_file} \
        -O ${recal_BAM_file} \
        --bqsr-recal-file ${BQSR_report}
```

-I : input BAM file.
-O : ouptut recalibrated BAM file.
--bqsr-recal-file : BQSR report file.

Q&A